Donghui Li, Ph.D.

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RESEARCH SUMMARY

I study hydroclimatic extremes in coupled natural and managed river systems. I develop physics-aware AI methods together with large-scale hydrological and hydrodynamic modeling frameworks to represent runoff, river routing, and water management. I connect these systems with Earth-system prediction ensembles to understand and predict hydroclimatic extremes across weather and climate timescales. Across this work, I integrate process-based modeling, high-performance computing, machine learning, and Earth observations.

RESEARCH INTERESTS

- Large-Scale Hydrological and Water Resources Modeling
- Flood, Drought, and Compound-Hazard Modeling
- Hydroclimatic Prediction and Projection Across Weather-to-Climate Timescales
- Physics-Aware AI for Earth and Environmental Systems

EDUCATION

2020.08–2024.08 **Ph.D. in Civil and Environmental Engineering; Minor in Statistics**
University of Illinois Urbana-Champaign, USA

2018.08–2020.08 **M.S. in Civil and Environmental Engineering**
University of Illinois Urbana-Champaign, USA

2014.09–2018.07 **B.E. in Hydraulic Engineering**
Tsinghua University, China

PROFESSIONAL EXPERIENCE

2024.09–present **Postdoctoral Research Associate**
Princeton University, USA; currently working with NOAA Geophysical Fluid Dynamics Laboratory

2025.01–2025.05 **Lecturer**
Princeton University, USA

2018.08–2024.08 **Graduate Research Assistant**
University of Illinois Urbana-Champaign, USA

SELECTED PROJECTS

Prediction of Hydroclimatic Extremes Across Timescales | Ongoing collaboration with NOAA GFDL

- Linking SPEAR seasonal prediction ensembles with distributed hydrological and hydrodynamic models to develop and evaluate seasonal flood-risk predictions.
- Investigating how prediction skill changes as atmospheric information propagates through land and river networks.

High-Resolution Flood Inundation Modeling | Project Lead | Industry-Sponsored Collaboration (*Confidential*)

- Led the proof-of-concept implementation of the coupled Tiger-HLM–TRITON framework at 30-meter resolution across the Delaware and Susquehanna River Basins.
- Drove the modeling chain with atmospheric forcings from a large climate ensemble to produce probabilistic inundation estimates for catastrophe-risk assessment.
- Delivered a technical report and probabilistic inundation data products to the sponsor.

Large-Scale Modeling of Reservoir Operations and Managed River Systems

- Developed a family of interpretable and transferable machine-learning models, GDROM and MODROM, to infer reservoir operating behavior and extend its representation to data-scarce reservoirs.
- Integrated these models offline with the National Water Model and the Variable Infiltration Capacity model to quantify how reservoir operations affect regulated streamflow, drought propagation, and water availability.
- Developed TigeRes, a two-way coupled reservoir–hydrology framework that integrates MODROM into Tiger-HLM.

PUBLICATIONS

Published

- Pu, T., Villarini, G., Amorim, R., **Li, D.**, Gangrade, S., Durr, M., & Janssen, B. (2026). Flood Responses to Climate Warming on Tribal Lands Using High-Resolution Hydrodynamic Modeling. *Geophysical Research Letters*. doi:10.1029/2026GL124766

- **Li, D.**, Michalek, A., & Villarini, G. (2026). Assessing the global performance of a parsimonious soil temperature model for frozen ground prediction. *Journal of Hydrology*. doi:10.1016/j.jhydrol.2026.135277
- **Li, D.**, & Villarini, G. (2026). Parsimonious and transferrable parameterization of reservoir operations: A modular approach for large-scale modeling. *Journal of Advances in Modeling Earth Systems*, *18*(2). doi:10.1029/2025MS005180
- Shi, H., Cai, X., Hu, X., Jamal, A., **Li, D.**, Sun, C., & Liang, X. Z. (2025). A sub-seasonal to seasonal climate forecast informed irrigation scheduling tool for the Contiguous United States. *Environmental Modelling & Software*, 106819. doi:10.1016/j.envsoft.2025.106819
- **Li, D.**[*], Vora, A.[*], & Cai, X. (2025). Streamflow simulation improvements enabled by a state-of-the-art algorithm for reservoir routing in the US National Water Model. *JAWRA Journal of the American Water Resources Association*. doi:10.1111/1752-1688.13244
- **Li, D.**[*], Chen, Y.[*], Lyu, L., & Cai, X. (2024). Uncovering historical reservoir operation rules and patterns: Insights from 452 large reservoirs in the contiguous United States. *Water Resources Research*. doi:10.1029/2023WR036686
- Vora, A., Cai, X., Chen, Y., & **Li, D.** (2024). Coupling reservoir operation and rainfall-runoff processes for streamflow simulation in watersheds. *Water Resources Research*. doi:10.1029/2023WR035703
- Chen, Y.[*], **Li, D.**[*], Zhao, Q., & Cai, X. (2022). Developing a generic data-driven reservoir operation model. *Advances in Water Resources*. doi:10.1016/j.advwatres.2022.104274
- Zhao, Q.[*], **Li, D.**[*], & Cai, X. (2021). Online generic diagnostic reservoir operation tools. *Environmental Modelling & Software*. doi:10.1016/j.envsoft.2020.104918

Under Review

- Cai, X., Chen, Y., Vora, A., **Li, D.**, Zheng, Z., Houtven, G., & Delaney, C. (Revised & Resubmitted). Forecast-Informed Reservoir Operation and Forecast Improvement with Reservoir Modification: A Perspective from Research to Practice in the United States. *Journal of Water Resources Planning and Management*.
- Michalek, A., **Li, D.**, Pu, T., & Villarini, G. (Revised & Resubmitted). Machine learning-based estimation of velocity-discharge power law parameters for ungauged locations. *Journal of Advances in Modeling Earth Systems*.
- Binjolkar, M., Michalek, A., Amorim, R., **Li, D.**, Maebius, S., Pu, T., Biswas, A., Ethier, S., & Villarini, G. (2026). Tiger-HLM: A GPU-Accelerated Distributed Hydrologic Model for High-Resolution Streamflow Simulation Across Continental Domains. *Advances in Water Resources*.
- Michalek, A., Binjolkar, M., Villarini, G., Amorim, R., **Li, D.**, & Pu, T. (2026). Tiger-HLM for Continental Hydrology: A CONUS-Wide Evaluation of Streamflow Generation Under Multiple Precipitation Forcings. *Advances in Water Resources*.

In Preparation

- **Li, D.**, Michalek, A., Amorim, R., & Villarini, G. (2026). TigeRes: Integrating Reservoir Operations into the Tiger-Hillslope Link Model.
- **Li, D.**, & Cai, X. (2026). Understanding dual effects of reservoir operation on hydrological drought and water supply drought.
- Pu, T., **Li, D.**, & Villarini, G. (2026). Terrain-Prior Attention Enables Event-Conditioned Urban Flood Mapping from Meteorological Forcing.
- Michalek, A., Amorim, R., **Li, D.**, Binjolkar, M., Pu, T., Villarini, G., Done, J. M., & Gangrade, S. (2026). High-resolution flood inundation maps across the contiguous United States under historical and pseudo-global warming scenarios.

[*] These authors contributed equally.

SOFTWARE AND DATA

GDROM | Co-Developer | Project link

- Machine-learning-based reservoir operation rules for 450+ CONUS reservoirs.
- Co-developed the machine-learning algorithm and data pipeline.

Tiger-HLM | Contributor | GitHub

- A high-performance, process-based hydrological model (rainfall–runoff–routing) accelerated using GPU (CUDA) and OpenMP.
- Implemented snow and soil-temperature modules for improved cold-region representation.
- Implemented an I/O module supporting large-scale simulations.

TigeRes | Developer | GitHub

- A version of the Tiger-HLM framework that includes coupled water management for regulated-streamflow simulation.

CONFERENCE PRESENTATIONS

- **Li, D.** & Villarini, G. (Dec. 2025). Parsimonious and Transferrable Parameterization of Reservoir Operations: A Modular Approach for Large-Scale Modeling. *AGU 2025* [poster].
- **Li, D.**, Michalek, A., & Villarini, G. (Dec. 2025). Global Products for Cold-Region Hydrological Modeling: Precipitation Phase

Partitioning and Frozen Ground Status Prediction. *AGU 2025* [poster].

- Pu, T., Villarini, G., Amorim, R., **Li, D.**, Gangrade, S., Durr, M., & Janssen, B. (Dec. 2025). High-Resolution Flood Inundation Mapping of U.S. Tribal Nations Under Current and Future Conditions: The Case of the Santee Sioux Nation. *AGU 2025* [poster; presented by Pu, T.].
- **Li, D.** & Cai, X. (Dec. 2024). Water Supply Drought Under Reservoir Regulation: Understanding Impacts and Developing an Integrated Drought Index. *AGU 2024* [poster].
- Barresi Armoa, O. L., Arnold, J., **Li, D.**, Chen, Y., Cai, X., Sauvage, S., & Sánchez-Pérez, J. M. (Jul. 2024). Development of reservoir operation functions in SWAT+ using a generic data-driven reservoir operation model. *2024 Strasbourg, France SWAT Conference* [oral; presented by Barresi Armoa, O. L.].
- Vora, A., **Li, D.**, Chen, Y., & Cai, X. (Jun. 2024). Changes in Regional Water Availability Resulting from Human Interferences in Hydrology. *CIROH 2024* [poster; presented by Vora, A.].
- Vora, A., Cai, X., Chen, Y., & **Li, D.** (May 2024). Regional Water Availability Changes Resulting from Human Interferences in Hydrology. *EWRI 2024* [oral; presented by Vora, A.].
- **Li, D.**, Chen, Y., & Cai, X. (Dec. 2023). Understanding Real-World Reservoir Operation with a Data-Driven Approach for Enhanced Hydrological and Water Resources Simulations. *AGU 2023* [poster].
- Vora, A., Cai, X., Chen, Y., & **Li, D.** (Dec. 2023). Coupling Reservoir Operation with Rainfall–Runoff Processes to Simulate Streamflow in Watersheds. *AGU 2023* [poster; presented by Vora, A.].
- **Li, D.**, Chen, Y., Zhao, Q., & Cai, X. (May 2023). Operation Rules and Patterns for 450+ Large Reservoirs in the Contiguous United States. *EWRI 2023* [oral].
- **Li, D.**, Chen, Y., & Cai, X. (Feb. 2023). Data-Driven Operation Rules for Reservoirs Across the CONUS. *USACE R&D Day 2023 at UIUC* [poster].
- **Li, D.**, Chen, Y., Zhao, Q., & Cai, X. (May 2022). Improving the Human Dimension of Hydrological Simulation Based on a Data-Driven Reservoir Operation Model. *EWRI 2022* [oral].
- Chen, Y., **Li, D.**, Zhao, Q., & Cai, X. (May 2022). Developing a Generic Data-Driven Reservoir Operation Model. *EWRI 2022* [oral].
- **Li, D.**, Zhao, Q., & Cai, X. (May 2021). DROT—A Diagnostic Reservoir Operation Tool. *EWRI 2021* [oral].
- **Li, D.**, Zhao, Q., & Cai, X. (May 2019). Decision Support Tool for Reservoir Operation Based on Derived Rules. *EWRI 2019* [oral].

AWARDS & FELLOWSHIPS

2023	**Mavis Future Faculty Fellowship** *University of Illinois Urbana-Champaign*; "to facilitate the training of the next generation of great engineering faculty."
2021 & 2023	**Conference Travel/Presentation Award** *University of Illinois Urbana-Champaign*
2018	**Yen Fellowship** *University of Illinois Urbana-Champaign*
2016 & 2017	**Academic Excellence Scholarship** *Tsinghua University*

TEACHING EXPERIENCE

Lecturer, High Meadows Environmental Institute, *Princeton University* 2025 Spring
ENV 423 Hydroclimatology

- Co-developed and delivered course materials for ENV 423 Hydroclimatology with Professor Villarini.
- Lectured on Python programming and its applications in hydroclimatology and statistical hydrology.

Wintersession Co-Facilitator, *Princeton University* 2025 Wintersession
Introduction to Physical Hydrological Modeling

- Lectured on the fundamentals of physically based hydrological modeling.
- Led a hands-on training workshop on the SWAT model.

Computing Teaching Assistant, *University of Illinois Urbana-Champaign* 2024 Spring
Computing-intensive courses across the CEE curriculum

- Led weekly seminars and tutorials focused on scientific programming.
- Held regular office hours providing technical assistance with programming problems.

Teaching Assistant, *University of Illinois Urbana-Champaign* 2021–2023 Fall
CEE 434 Environmental Systems I

- Lectured on the fundamentals of machine learning and its applications to water resources engineering.
- Graded assignments and held office hours.

TECHNICAL SKILLS

Programming & Tools: Python, R, C/C++, Fortran, MATLAB, SQL, Git, scikit-learn, PyTorch, LaTeX
Environmental Modeling: VIC, WRF-Hydro, SWAT, TRITON, Tiger-HLM
High-Performance Computing: UNIX/Linux, Bash scripting, OpenMP/MPI, CUDA, Kokkos

ACADEMIC AND VOLUNTEER SERVICE

- Journal Reviewer: *Earth's Future*; *Ecological Indicators*; *Environmental Modelling & Software*; *Geoscientific Model Development*; *Hydrology and Earth System Sciences*; *Journal of Hydrology*; *Journal of the American Water Resources Association*; *Water Resources Research*.
- Early Career Convener, AGU 2025: "H004—Advancements in Physics-Based, Integrated Hydrologic Modeling to Support Water Resources Management."
- Session Moderator, Environmental & Water Resources Institute (EWRI) Conference, 2022.

PROFESSIONAL AFFILIATIONS

- Member, American Geophysical Union (AGU)
- Member, American Meteorological Society (AMS)
- Member, American Society of Civil Engineers (ASCE)